

The 3/16 Illusion: From Selberg to Random Hyperbolic Surfaces

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Let $X = \Gamma \backslash \mathbb{H}^2$ be a compact hyperbolic surface, with Laplace–Beltrami spectrum

$$0 = \lambda_0(X) < \lambda_1(X) \leq \lambda_2(X) \leq \cdots.$$

The first positive eigenvalue λ_1 , the spectral gap, measures a form of global connectivity: a small gap is associated with bottlenecks and slow mixing, while a large gap indicates that the surface is difficult to separate into macroscopic pieces. The natural reference scale is $1/4$: the spectrum of the hyperbolic plane is $[1/4, \infty)$, and Huber proved that in the large-genus regime

$$\limsup_{g \rightarrow \infty} \sup_{X \in \mathcal{M}_g} \lambda_1(X) \leq \frac{1}{4}.$$

Thus $1/4$ is the asymptotically maximal possible gap for large compact hyperbolic surfaces [9, 8].

The arithmetic story begins with Selberg. In 1965 he conjectured that for congruence covers of the modular surface one should have

$$\lambda_1 \geq \frac{1}{4},$$

the celebrated Selberg eigenvalue conjecture. In the same work he proved the weaker bound $3/16$. Decades of progress improved the constant, culminating in the bound

$$\lambda_1 \geq \frac{975}{4096} \approx 0.2380,$$

due to Kim and Sarnak [10, 5]. The gap between $975/4096$ and $1/4$ is small numerically but remains a major open problem in the arithmetic setting.

At first sight, one might therefore regard $3/16$ as a deep threshold. The random-surface story suggests otherwise. For Weil–Petersson random surfaces of genus g , Wu–Xue and Lipnowski–Wright proved independently that, with probability tending to one,

$$\lambda_1(X) \geq \frac{3}{16} - \varepsilon.$$

The same number also appeared for random covers of compact hyperbolic surfaces in work of Magee, Naud and Puder [11, 6, 7]. In this setting, however, subsequent work moved beyond $3/16$: Hide and Magee proved that random covers of finite-area hyperbolic surfaces have no new eigenvalues below $1/4 - \varepsilon$, and obtained closed surfaces with first eigenvalue tending to $1/4$ [4].

The analogy with random regular graphs is especially revealing. Friedman’s theorem, following Alon’s conjecture, shows that a large random regular graph has essentially the optimal nontrivial spectral edge predicted by its universal covering tree [3]. The corresponding universal cover for a hyperbolic surface is $\mathbb{H}^2$, whose spectral edge is exactly $1/4$. This suggests a useful heuristic: a sufficiently typical large hyperbolic surface should inherit the spectral edge of its universal cover.

Anantharaman and Monk made this heuristic precise in a sequence of works. Their 2024 paper gave a roadmap toward the optimal result; their subsequent analysis first reached $2/9 - \varepsilon$ and then, in joint work presented in 2026, established that for every $\varepsilon > 0$,

$$\mathbb{P}_g^{\text{WP}} \left(\lambda_1(X) \geq \frac{1}{4} - \varepsilon \right) \rightarrow 1 \quad (g \rightarrow \infty).$$

The intermediate values are instructive: the trace-method analysis gives $3/16$ at a first level of precision, $2/9$ after a finer treatment of the relevant geometric contributions, and ultimately reaches $1/4$ after controlling the expansion much more deeply [1, 2, 8].

This history changes how $3/16$ should be interpreted. In the random setting, it is not a universal obstruction: it marks a point at which an available method becomes insufficiently precise, rather than a spectral wall that the geometry itself cannot cross. The number $1/4$ is more naturally viewed as a universal-cover edge, in the spirit of a Ramanujan bound. From this perspective, the interesting question is not why random surfaces stop at $3/16$, but why additional cancellation, exclusion of local “tangles,” and increasingly precise trace estimates allow the random model to approach the edge $1/4$.

This also clarifies why the arithmetic problem does not disappear. Congruence covers are highly structured, rigid, and arithmetically correlated; they are not samples from the Weil–Petersson measure. As Monk emphasizes, the two settings are essentially orthogonal: the random argument averages over moduli space, whereas Selberg’s conjecture concerns a thin, highly structured arithmetic family [8]. Consequently, the near-optimal random result does not imply the arithmetic conjecture.

Working hypothesis. My interpretation is that the apparent $3/16$ barrier in the random theory is primarily methodological rather than spectral: as the trace analysis is pushed to finer orders, non-generic geometric configurations can be increasingly excluded and the spectrum moves toward the universal-cover edge $1/4$. This is a working hypothesis/interpretation, not a new theorem; the corresponding difficulty for arithmetic congruence families remains open.

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